2 **C**

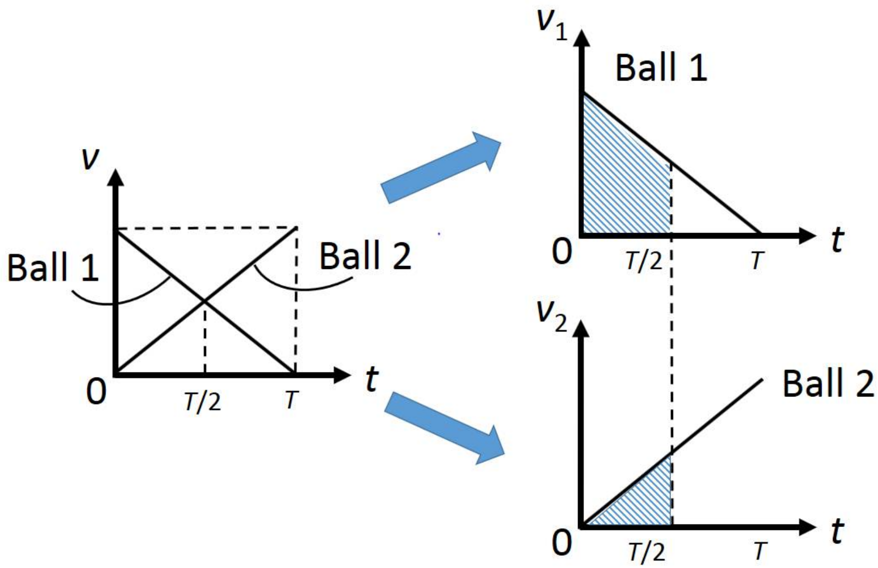
Method 1: Faster (guess & check using graphical method)

If Ball 1 doesn't collide with Ball 2, it will take time T to reach the top. Similarly, Ball 2 will take time T to reach the bottom.

Because both balls are experiencing the same gravitational acceleration ($g \sin\theta$), the initial speed of Ball 1 is the same as the final speed of Ball 2.

Guess: Is it possible that the balls collide at time = $T/2$?

Check: At time = $T/2$, apply the math of similar triangles to see that Ball 1 travels ¾ of the way up the slope (refer to the shaded area under Ball 1's curve). At time = $T/2$, Ball 2 travels ¼ of the way down the slope (refer to the shaded area under Ball 2's curve). So the balls do indeed collide at time = $T/2$, and Ball 1 travels 3 times further than Ball 2.



Method 2: Slower (using equations and graphs of kinematics)

Apply "$s = ut + (1/2)\, at^2$" and let t be the time of collision:

For Ball 1: $$s_{AC} = ut + \frac{1}{2}(-a)t^2 = ut - \frac{1}{2}at^2$$

For Ball 2: $s_{BC} = (0)t + \frac{1}{2}at^2 = \frac{1}{2}at^2$

$s_{AB} = s_{AC} + s_{BC} = ut - \frac{1}{2}at^2 + \frac{1}{2}at^2$

$\therefore s_{AB} = ut$ --- (1)

Suppose there is no collision. Let *T* be the time taken by either ball to travel the entire length of the slope s_{AB}.

Apply "$s = \left(\frac{u+v}{2}\right)t$" to Ball 1 gives: $s_{AB} = \left(\frac{u+0}{2}\right)T$

$\therefore s_{AB} = \frac{uT}{2}$ --- (2)

Equating (1) and (2) gives $t = \frac{T}{2}$

Refer to the previous *v-t* graphs for Balls 1 & 2. After time *T*/2, apply the math of similar triangles to see that Ball 1 travels ¾ of the way up the slope (refer to the shaded area under Ball 1's curve). After *T*/2, Ball 2 travels ¼ of the way down the slope (refer to the shaded area under Ball 2's curve). So the balls collide after *T*/2, and Ball 1 travels 3 times further than Ball 2.

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